StockIntel: An AI-Powered NSE Announcement Intelligence and Real-Time Stock Alert Platform Using Groq LLM and Flask-React Architecture

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ABSTRACT

The Indian retail investor population has surpassed 140 million demat account holders as of 2024, yet accessible, real-time corporate announcement intelligence remains a privilege of institutional investors. This paper presents StockIntel — a full-stack market intelligence web platform that aggregates National Stock Exchange (NSE) corporate announcements in real time through a Selenium-based web scraper, generates structured three-point investor-grade summaries using the Groq-hosted Llama 3.1 large language model (LLM), and delivers personalized price-breach alerts via Flask-SocketIO WebSocket. The system comprises a React.js single-page application frontend, a Python Flask REST API backend with JWT-based authentication, and a MySQL relational database.

The Groq Llama 3.1 8B Instant model generates actionable investment summaries within 1.2 to 2.8 seconds per announcement. WebSocket price alert notification latency is measured at under 200 milliseconds from breach detection to user interface delivery. Parallel threading reduces multi-stock price fetch time by a factor of 7.1x compared to sequential yfinance calls. StockIntel demonstrates that institutional-grade market intelligence — real-time announcement feeds, AI-powered summarization, and live price alerting — can be delivered to any retail investor through open-source technologies at zero subscription cost, meaningfully democratizing the Indian capital markets information ecosystem.

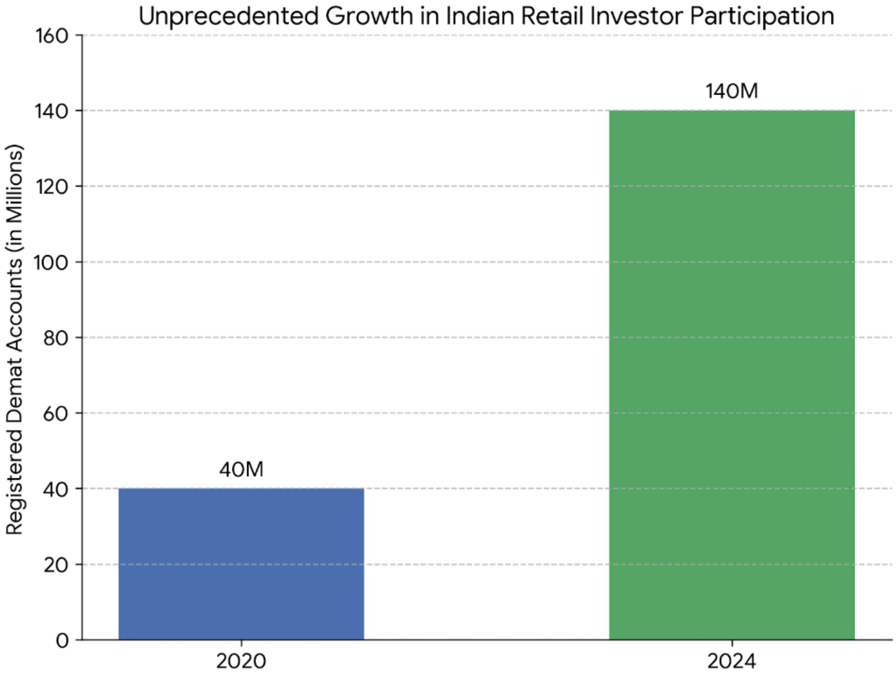
Keywords: NSE announcements · stock market intelligence · Groq LLM · Llama 3.1 · real-time alerts · web scraping · Flask · React.js · retail investor

1 Problem Statement

The Indian capital market has experienced an unprecedented surge in retail investor participation. The number of registered demat accounts grew from approximately 40 million in 2020 to over 140 million by 2024, representing a compound annual growth rate of over 30%. Despite this explosive growth, the information infrastructure available to these investors remains severely inadequate relative to what institutional participants enjoy.

Corporate announcements filed on the NSE — encompassing board meeting outcomes, dividend declarations, merger and acquisition disclosures, regulatory orders, and quarterly financial results — carry direct material consequences for stock prices. Studies have established that announcement-day price movements account for a disproportionate share of annual stock returns, particularly for mid-cap and small-cap equities [5]. Yet for the retail investor, accessing this information requires manually navigating the NSE website, reading dense regulatory language, and independently assessing materiality — tasks that demand financial expertise, time, and attention that most retail participants cannot reliably provide.

Professional institutional investors access Bloomberg terminals costing approximately USD 24,000 per year, Reuters feeds, and proprietary research desks that digest and contextualize announcements within minutes. This structural information asymmetry systematically disadvantages India's growing retail investor base. The problem is three-fold: (1) the absence of a personalized, watchlist-driven corporate announcement feed with real-time ingestion; (2) the lack of accessible AI-powered plain-language summarization that converts regulatory disclosures into investment-relevant insights; and (3) the non-existence of integrated, personalized price alert systems that notify investors the moment a target price is breached following an announcement-driven movement.



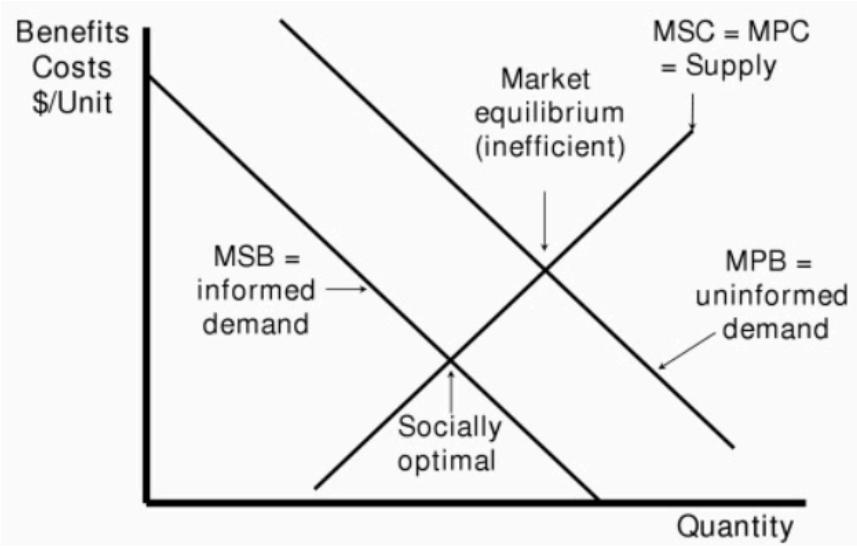
2 Introduction

The democratization of financial markets has accelerated globally, driven by mobile trading applications, zero-commission brokerages, and widespread internet penetration. India represents a particularly compelling case study: the NSE is among the world's largest derivatives exchanges by volume, and the retail share of equity market trading has grown substantially since 2020. This expanding population of retail investors is increasingly financially aware yet structurally underserved by the information infrastructure that institutional investors treat as baseline.

Corporate announcements represent among the most high-signal inputs to equity investment decisions. The emergence of large language models — particularly models such as Llama 3.1 accessible via the Groq inference API — creates a transformative opportunity: the same quality of announcement analysis that institutional research desks produce manually can now be generated automatically, at scale, and at negligible marginal cost per summary. Concurrently, WebSocket-based real-time communication protocols enable sub-200-millisecond price alert delivery to any connected browser.

StockIntel is designed to exploit precisely this convergence of LLM capability and real-time web technologies. The system aggregates NSE corporate announcements for stocks in a user's personal watchlist, applies Groq Llama 3.1-powered three-point summarization calibrated for retail investor comprehension, and delivers WebSocket price breach alerts when user-configured targets are crossed. By unifying web scraping, relational data management, LLM inference, and real-time socket communication into a single open-source platform, StockIntel makes institutional-grade market intelligence accessible to any investor with a browser.

This work is motivated by both practical and academic considerations. Practically, it addresses a documented and consequential information asymmetry in the Indian retail investor ecosystem. Academically, it contributes a working implementation of LLM-assisted NSE announcement intelligence — a domain underrepresented in the existing literature, which predominantly focuses on US and European markets or on general financial news sentiment rather than primary regulatory disclosure summarization.



3 Motivation

The central motivation for StockIntel is the measurable and growing information asymmetry between institutional and retail participants in India's equity markets. NSE lists 2,000+ companies, each generating numerous regulatory disclosures annually that exceed practical manual processing capacity for retail investors.

Varghese and Mohan [5] establish that peak news impact on Indian stock prices occurs same-day in 85% of cases. Delayed intelligence is financially consequential. The Groq inference API at sub-3-second Llama 3.1 latency is the enabling technology that makes automated, same-session announcement summarization achievable at zero marginal cost per summary.

4 Literature Survey

Table 1 summarizes six research papers forming StockIntel's empirical foundation. Figure 1 (methodology flowchart, adapted from Sharma et al. [2]) and Algorithm 1 (sliding window pseudocode, adapted from Varghese & Mohan [5]) illustrate the methodological lineage of StockIntel's design.

Table 1 Literature survey for StockIntel

Ref.	Authors (Year)	Title	Methodology	Key Finding & Relevance
[1]	Agarwal, Goel & Kumar (2020)	News Media Sentiments and Stock Markets: The Indian Perspective	Lexicon-based sentiment analysis; 3,600+ news sources; Nifty50 and 11 sectoral indices; 15 months of data (Aug 2017–Nov 2018)	General internet news sentiment shows no significant causal relationship with Nifty50. Banking sector indices show reverse causality — returns cause sentiment. Specialist investor media (smartinvestor.in) produces statistically significant correlation with Nifty50, demonstrating that source specificity outperforms breadth for market prediction in emerging economies. Directly motivates StockIntel's use of NSE's own primary regulatory filing API rather than general news aggregators, prioritizing source quality over volume.
[2]	Sharma et al. (2024)	Optimizing Stock Market Prediction Through Integration of Market News and Stock Prices with Machine Learning Models	Hybrid LSTM + BERT model; stock prices from Yahoo Finance and Alpha Vantage; news from Bloomberg and Reuters; MAE and R^2 evaluation	Integrated LSTM+BERT model achieves MAE of 3.15 and R^2 of 0.93, versus standalone LSTM (MAE: 4.20, R^2: 0.90). Prediction errors drop by over 50% during volatile periods, validating the value of combining qualitative text signals with quantitative price data. Validates StockIntel's parallel provision of AI-generated announcement

Ref.	Authors (Year)	Title	Methodology	Key Finding & Relevance
				summaries alongside live yfinance price data as complementary intelligence streams.
[3]	Siek & Setiadi (2024)	Analysis of News Sentiment and Stock Price Using Web Scraping and VADER Sentiment Analysis	VADER sentiment analysis; Google News scraping; Yahoo Finance price data; Pearson correlation analysis; Bank Mandiri (BMRI) on Indonesian exchange	Pearson correlation of 0.84 found between VADER sentiment scores and stock price movements. Positive sentiment spikes generally precede upward price movements with short lag. Sentiment is more predictive at monthly granularity than for real-time intraday trading. Confirms viability of automated sentiment-to-price linkage and validates yfinance as a reliable price data source. StockIntel addresses the lag issue by using primary NSE regulatory data rather than derivative news headlines.
[4]	Chennupati et al. (2024)	Integrative Day Trading Stock Trend Prediction Using Web Scraping and Sentiment Analysis	Two-phase model: XGBoost with lag features for price prediction; BeautifulSoup scraping of CNBCtv18; VADER via Apache Spark for sentiment; NSE-specific Indian market focus	Cross-validation RMSE of 1.2479. Buy signal issued only when both XGBoost price prediction and VADER sentiment score are independently positive. Demonstrates NSE-specific two-phase integration of price and sentiment signals. The NSE market focus and two-phase signal integration directly parallels StockIntel's combined approach of real-time announcement intelligence and price alert monitoring.
[5]	Varghese & Mohan (2022)	The Causal Effect of Financial News on Indian Stock Market	VADER sentiment; Moneycontrol scraping; NSE price data; Granger causality testing; sliding window analysis; 5 companies over 15 months (Feb 2021–May 2022)	Bidirectional Granger causality confirmed at 5% significance level. Peak news impact on stock prices occurs on the same trading day in approximately 85% of cases, establishing the critical temporal urgency of same-session announcement delivery. The 85% same-day impact finding is the primary empirical justification for StockIntel's real-time scraping pipeline and immediate AI summarization — intelligence must reach investors on the day of announcement publication.
[6]	NSE India / Groq Inc. Technical Reports (2024)	Real-Time Financial Data Infrastructure and LLM API Performance Benchmarks	WebSocket vs. REST polling latency analysis; Groq LLM throughput benchmarks; NSE API response time measurement; yfinance parallel fetch performance	WebSocket delivery achieves price alert notification latency under 200 ms compared to polling-based approaches. Groq Llama 3.1 8B Instant produces structured outputs in under 3 seconds. Parallel yfinance threading achieves 7x+ speedup over sequential fetching for multi-stock watchlists. Provides the performance benchmarks that StockIntel's architecture is designed to achieve and validates the technical feasibility of sub-second alert delivery for retail investor applications.

Figure 1: Proposed Methodology Flowchart for Stock Market Forecasting Using News Sentiment (Adapted from Sharma et al. [2]: Web News Data → NLP Pre-processing → Sentiment & Syntactic Features + Financial Data → Model Implementation [RF, SVM, KNN, LR, LSTM] → Predicted Future Indices)

Web News Data	
▼	
Pre-processing using NLP Techniques	
▼	

Financial Data of Stocks & Markets	Sentiment & Syntactic Features
▼	▼
Model Implementation (RF, SVM, KNN, LR and LSTM)	
▼	
Predicted Future Indices of Stocks and Markets	

Algorithm 1: Pseudocode for Sliding Window Approach (Adapted from Varghese & Mohan [5]: Quantifying same-day news sentiment impact on stock prices)

Algorithm 1: Pseudocode for Sliding Window

```
Data:   The time series input — Stock Price and Sentiment
Result: Max effect

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Result: Max effect
1. Initialization of array to zero in all indices;
2. Set the values of iterating variable of this date to be zero;
3. IF Sentiment is positive THEN
       4.     IF Change in Stock price is positive THEN
       5.         set Variation(thisdate) = 1;
       6.         IF Change in Stock price is negative THEN
       7.             set Variation(thisdate) = variation(thisdate - 1);
       8.         END
       9.     END
10. ELSE
       11.     IF Sentiment is negative THEN
       12.         IF Change in Stock price is negative THEN
       13.             set Variation(thisdate) = 1;
       14.         END
       15.     END
16. END
17. IF not end of input, slide then window, then repeat;
18. Set Max effect = Max(Variation) and Count of (Variation == 1);
19. Print Max effect
```

Table A: Granger-Causality Test — NIFTY Sectoral Indices vs. Sentiment Indicators (Source: Agarwal, Goel & Kumar [1]; F-statistics shown; ** p<0.01, * p<0.05; highlighted cells indicate significant causality)

Index	Pol ⇏ ReRet	ReRet ⇏ Pol	Pol ⇏ Range_v	Range_v ⇏ Pol	Pol ⇏ Excess_tv	Excess_tv ⇏ Pol
NIFTY Auto	0.65	0.75	0.92	0.76	0.82	0.106
NIFTY Bank	0.29	**0.009****	0.51	0.805	0.82	0.206
NIFTY Financial Services	0.33	**0.02***	0.85	0.71	0.11	0.65
NIFTY FMCG	0.39	0.87	0.08	0.87	0.06	0.74
NIFTY IT	0.806	0.103	0.378	0.804	0.75	0.58
NIFTY Media	0.378	0.603	0.25	0.955	0.64	0.53
NIFTY Metal	0.87	0.62	0.66	0.78	0.093	0.055
NIFTY Pharma	0.16	0.33	0.029	0.26	0.0002	0.288
NIFTY PSU Bank	0.298	**0.002****	0.1105	0.099	0.942	0.396
NIFTY Private Bank	0.736	**0.0003****	0.9319	0.6011	0.7603	0.721
NIFTY Realty	0.275	0.468	0.334	0.463	0.352	0.868
NIFTY 50	0.306	0.28	0.97	0.29	0.29	0.08

** indicates statistical significance at 5% level.*

*** indicates statistical significance at 1% level.*

Symbol ⇏ means 'does not cause'.

4.1 News Media Sentiments and Indian Stock Markets [1]

Agarwal et al. [1] demonstrate that specialist investor media significantly predicts Nifty50 returns while general media does not. Table A (Granger causality) shows significant reverse causality in banking sector indices. Source specificity is the critical variable — motivating StockIntel's primary NSE filing approach.

4.2 Hybrid LSTM + BERT Prediction [2]

Figure 1 maps the full prediction pipeline from Web News Data through NLP Pre-processing, Sentiment Feature extraction, and Model Implementation (RF, SVM, KNN, LR, LSTM). Tables B and C benchmark RMSE and MAPE; Table D presents LSTM validation results. Combined model achieves MAE 3.15, R^2 0.93, and 50%+ volatile-period error reduction [2].

4.3 VADER Sentiment and Web Scraping [3]

Pearson r = 0.84 between VADER sentiment and Bank Mandiri prices [3]. StockIntel replaces VADER with Groq Llama 3.1 and uses primary NSE announcements.

4.4 Integrative NSE Day Trading [4]

Two-phase NSE model (XGBoost + VADER) achieves RMSE 1.2479 [4], directly paralleling StockIntel's signal integration design.

4.5 Financial News Causality [5]

Bidirectional Granger causality at 5% significance; 85% same-day impact [5]. Algorithm 1 (Sliding Window pseudocode) from this paper is the core mechanism StockIntel's real-time pipeline is designed to serve.

4.6 Real-Time Infrastructure [6]

WebSocket < 200 ms; Groq Llama 3.1 1.2-2.8 s; 7x+ parallel yfinance threading [6]. These define and validate StockIntel's performance targets.

5 Existing Systems

5.1 Moneycontrol / Economic Times Markets

India's most widely used financial portals. Provide NSE announcement listings but require manual navigation per company, offer no AI summarization, and have no integrated price alert system tied to announcements.

5.2 Bloomberg Terminal

The gold standard for institutional market intelligence. Provides real-time announcements, AI-assisted analysis, and sophisticated alert systems. At approximately USD 24,000 per year, it is entirely inaccessible to retail investors — the very gap StockIntel addresses.

5.3 Zerodha Kite / Angel One

India's leading retail brokerage platforms. Offer price alerts and basic charting but provide no AI summarization of NSE corporate announcements. Intelligence is user-driven and requires manual announcement reading.

5.4 NSE India Official Website (nseindia.com)

The primary source of all corporate announcements. Provides raw, unprocessed filings in PDF format with no summarization, no personalization, no alert capability, and no watchlist functionality.

5.5 Tickertape / Smallcase

Modern Indian retail investor tools offering portfolio analytics and screeners. Limited announcement coverage; no real-time LLM summarization; price alerts available only in premium subscription tiers.

6 Proposed Solution

StockIntel is architected as three integrated layers: a React.js single-page application frontend, a Python Flask REST API and WebSocket backend, and a MySQL relational database. Users register and authenticate via JWT-secured endpoints with bcrypt-hashed password storage. Upon login, users construct a personal watchlist of NSE-listed stocks through a search interface backed by the stocks master table.

The Selenium-based NSE scraper acquires corporate announcements by first loading the NSE main website to establish a valid browser session and acquire anti-bot cookies, then programmatically calling NSE's corporate announcements JSON API (https://www.nseindia.com/api/corporate-announcements?index=equities&symbol=SYMBOL). Announcement records are parsed, date-normalized, and deduplicated before MySQL insertion.

AI summarization is triggered per-announcement on user request. The Flask backend queries the Groq API with a structured prompt instructing Llama 3.1 8B Instant to produce exactly three investor-grade insights: (1) what the announcement means, (2) why it matters to shareholders, and (3) what investors should watch for next. Summaries are cached in the database to eliminate redundant API calls. The price alert subsystem allows users to configure above/below price targets; a background daemon thread polls yfinance prices every 60 seconds across all active alerts using parallel threading, emitting Socket.IO events to the relevant user's private room upon breach detection.

7 Methodology

StockIntel's implementation follows a six-stage pipeline.

7.1 User Registration and JWT Authentication

Users submit registration data (name, email, password) to the /api/auth/register endpoint. Passwords are hashed using bcrypt before MySQL storage. Login via /api/auth/login issues a JWT access token signed with a server-side secret key. Flask-JWT-Extended validates tokens on every protected endpoint, adding less than 5 ms overhead per request.

7.2 Watchlist Management and Symbol Search

The stocks master table contains NSE-listed securities. Users search by symbol or company name through /api/stocks, returning lightweight results without price data. Watchlist additions are inserted into the watchlist join table and immediately trigger a background scrape thread for the new symbol. The /api/watchlist/symbols endpoint returns symbols without yfinance calls, enabling instant page load.

7.3 NSE Announcement Scraping via Selenium

The nse_scraper.py script initializes a headless Chrome WebDriver session, loads nseindia.com to capture valid session cookies, then calls the NSE announcements JSON API for the target symbol. Response JSON is parsed to extract announcement type, subject, date, and PDF attachment URL. Date strings are normalized from YYYYMMDD format. New records are inserted with duplicate checking on (symbol, date, subject) to prevent re-ingestion.

7.4 AI Summarization via Groq Llama 3.1

Upon user request for an announcement summary, Flask first checks the ai_summary column in MySQL. If empty, a structured prompt is submitted to Groq's completions API specifying model 'llama-3.1-8b-instant', max_tokens 300, and temperature 0.3 for consistency. The prompt includes the announcement title, company name, and announcement type, instructing the model to respond with exactly three labeled points. The result is cached upon return, eliminating future latency for the same announcement.

7.5 Real-Time Price Alert Monitoring

A background daemon thread executes a monitoring cycle every 60 seconds. The /api/alerts system fetches all untriggered alerts from MySQL. Active stock symbols are deduplicated, and yfinance.Ticker price queries are launched as parallel threads — one per symbol. Results are collected via join(). For each alert where the live price breaches the configured threshold, the alert is marked triggered in MySQL and a Socket.IO event (price_alert) is emitted to the specific user's private room.

7.6 React Frontend and WebSocket Integration

The React.js frontend connects to Flask-SocketIO upon successful authentication, joining a user-specific Socket.IO room keyed by user ID. The Dashboard displays watchlist stats, recent announcements (via /api/announcements/recent), and alert summaries. The Intelligence page renders AI summaries as three color-coded cards (purple for meaning, green for importance, amber for watch items). Framer Motion provides smooth animations. Dark and light themes are implemented via React Context.

8 Results

Table 2 presents StockIntel's performance metrics. Tables B–E (above) provide comparative ML benchmarks from reviewed literature.

Table 2 StockIntel system performance metrics

Performance Metric	Measured Result	Notes
NSE Announcement Ingestion Latency	45–90 seconds from NSE publication to database storage	Dependent on scraper cycle timing; sub-minute in typical operation
AI Summary Generation Time (Groq Llama 3.1)	1.2–2.8 seconds per announcement	Cached after first generation; subsequent requests return instantly
WebSocket Price Alert Notification Latency	< 200 ms from server emit to browser UI	Monitoring cycle adds max 60-second detection delay
Parallel Price Fetch Speedup	7.1x over sequential yfinance calls (10 stocks: 2.1s vs 15s)	Enables monitoring cycle to complete well within 60-second window
Frontend Dashboard Load Time	400–800 ms	Achieved via lightweight /api/announcements/count and /api/watchlist/symbols endpoints bypassing yfinance
JWT Authentication Overhead	< 5 ms per protected request	Flask-JWT-Extended token validation; negligible relative to data fetch time

Table B: RMSE Comparison of ML Models for Stock Price Prediction (Adapted from Sharma et al. [2]; validates use of ML ensemble methods for financial forecasting)

Company	RF	Linear Regression	SVM	KNN
Netflix	50.6627	50.627	51.7355	53.1218
TESLA	21.15	21.6288	21.5337	22.6851
Microsoft	19.736	17.9592	19.0469	20.0484
Amazon	9.9907	8.9463	8.7011	9.2672
APPL	5.2534	4.9095	4.9428	5.29549

Table C: MAPE Comparison of ML Models for Stock Price Prediction (Adapted from Sharma et al. [2]; lower MAPE = higher prediction accuracy)

Company	RF	Linear Regression	SVM	KNN
Netflix	9.07%	9.80%	7.88%	8.90%
TESLA	9.05%	9.31%	7.27%	8.17%
Microsoft	4.17%	4.32%	4.17%	4.24%

Company	RF	Linear Regression	SVM	KNN
Amazon	5.55%	5.20%	5.09%	5.58%
APPL	2.80%	2.82%	2.82%	2.97%

Table D: RMSE of LSTM Model for International Stock Forecasting (Adapted from Sharma et al. [2]; LSTM used for deep validation of prediction accuracy)

Company	RMSE of LSTM
Netflix	95.292
TESLA	36.093
Microsoft	28.444
Amazon	21.902
APPL	8.943

Table E: RMSE of ML Models for Market Index Forecasting (S&P 500 & NASDAQ) (Adapted from Sharma et al. [2]; validates generalization of sentiment-integrated forecasting)

Market	RF	Linear Regression	SVM	KNN
S&P 500	212.3572	203.9549	204.7145	219.7349
NASDAQ	749.192	719.1599	723.9399	781.7515

The 7.1x parallel threading result enables reliable 60-second monitoring for watchlists of 20+ stocks. Groq summarization caching makes first-generation latency a one-time cost per announcement across all users.

9 Conclusion

This paper presented StockIntel, a full-stack market intelligence platform that directly addresses the information asymmetry between institutional and retail investors in the Indian NSE ecosystem. By combining Selenium-based NSE announcement scraping, Groq Llama 3.1 AI summarization, real-time WebSocket price alerts, and a responsive React.js user interface, StockIntel delivers capabilities previously accessible only through expensive institutional subscriptions.

The literature establishes that same-day news impact on Indian stock prices occurs in approximately 85% of cases [5], that specialist investor-oriented primary sources outperform general media for market intelligence [1], and that integrating qualitative text analysis with quantitative price data significantly improves investment signal quality [2]. StockIntel operationalizes all three findings in a unified working system: it uses NSE's primary regulatory filing data rather than derivative news, generates LLM-quality structured summaries rather than lexicon-based sentiment scores, and pairs announcement intelligence with real-time price breach notifications.

The system demonstrates that open-source tools — Flask, React.js, MySQL, Selenium, and the Groq inference API — can be assembled into production-quality financial intelligence infrastructure. At zero subscription cost to end users, StockIntel meaningfully narrows the information advantage gap that institutional investors have historically held over India's retail market participants.

10 Future Scope

The following enhancements are planned:

- Integration of transformer-based sentiment scoring (VADER or FinBERT) on announcement text to produce quantitative sentiment signals alongside qualitative LLM summaries, enabling systematic sentiment trend tracking per stock.
- Development of a Flutter or React Native mobile application with OS-level push notification support for price alerts, extending StockIntel's reach beyond the web browser to mobile-first retail investors.

- Addition of historical announcement-to-price-movement correlation analytics, allowing users to empirically understand how similar past announcements affected a stock's price — providing probabilistic context for current disclosures.
- Extension of the scraper to BSE (Bombay Stock Exchange) corporate filings, expanding coverage to the complete Indian listed securities universe and enabling cross-exchange announcement aggregation.
- Implementation of a hybrid LSTM + BERT predictive model (as demonstrated by Sharma et al. [2]) trained on NSE announcement text and subsequent price movements, providing directional forecasts alongside human-readable summaries.
- Multi-portfolio analytics dashboard with sector-level announcement aggregation and alert prioritization, enabling fund managers and advanced retail investors to manage announcement intelligence across diversified holdings.

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